

Analysis of Linear Programming

Part 3a: Preliminaries

IE 311 – Operations Research I

An optimization problem with linear relations between continuous variables is a **Linear Program**:

$$\max; \sum_{j=1}^n c_j x_j \quad \text{s.t.} \quad \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m, \quad x_j \geq 0, \quad j = 1, \dots, n.$$

More generally, any Linear Program can be written in **Standard Form** (how?):

$$\min; c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0.$$

How to Transform any LP to Standard Form?

The standard form has only **equality constraints** and **nonnegative variables**.

- For each $\leq$ constraint, add a nonnegative **slack** variable:
 $a^\top x \leq b \Rightarrow a^\top x + s = b, s \geq 0.$
- For each $\geq$ constraint, subtract a nonnegative **surplus** variable:
 $a^\top x \geq b \Rightarrow a^\top x - s = b, s \geq 0.$
- For each **nonpositive** variable x , substitute a new nonnegative variable x' and replace x with $-x'$.
- For each **unrestricted (free)** variable x , substitute two new nonnegative variables x^+, x^- and replace x with $x^+ - x^-$.

We will analyze LPs in four main steps:

- **Preliminaries:** linear algebra and convexity review, basics of polyhedra.
- **Solution Methods:** the simplex method and its variants.
- **Theory:** duality.
- **Interpretations:** sensitivity and post-optimality analysis.

We will first briefly review the following concepts:

- Matrices / Vectors
- Linear Independence / Rank
- Linear Systems
- Convex Functions
- Convex Sets
- Hyperplane / Half-space

Then we will start analyzing the properties of **polyhedra**, the feasible region of LPs.

• A **matrix** is any rectangular array of numbers.

• Examples: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\begin{bmatrix} 1 & -2 & 4 \\ -4 & 8 & 7 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $[3 \ 4 \ -2]$

• In general, if a matrix A has m rows and n columns, we call it an $m \times n$ matrix (denoted by $A_{m \times n}$). The (i, j) entry of a matrix A is denoted by a_{ij} .

• A **row vector** is a matrix with a single row.

• Example: $[3 \ 4 \ -2]$

• A **column vector** is a matrix with a single column.

• Example: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

If u and v are n -vectors, their **inner (dot, scalar) product** is $u \cdot v = \sum_{i=1}^n u_i v_i$.

If A is an $m \times n$ matrix and B an $n \times p$ matrix, the **matrix product** $C = AB$ is a well-defined $m \times p$ matrix with entries

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}, \quad i = 1, \dots, m, \quad j = 1, \dots, p.$$

If u and v are n -column vectors, then $u \cdot v = u^{\top} v = v^{\top} u$.

- **Identity Matrix:** A square matrix with all zero entries except the diagonal entries which are 1.

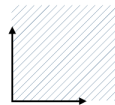
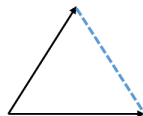
$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

- **Unit Vectors:** Columns of the identity matrix.

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

- **Zero Matrix/Vector:** A matrix/vector with all zero entries.

- Given vectors $v_1, v_2, \dots, v_k$ (of same size), any vector of the form $a_1 v_1 + a_2 v_2 + \dots + a_k v_k$ is called
 - a **linear combination** for arbitrary scalar $a_1, a_2, \dots, a_k$,
 - a **convex combination** for nonnegative scalars $a_1, a_2, \dots, a_k$ satisfying $a_1 + a_2 + \dots + a_k = 1$,
 - a **conic combination** for nonnegative scalar $a_1, a_2, \dots, a_k$.



- A set of m -dimensional vectors $\{v_1, v_2, \dots, v_k\}$ is called **linearly independent** if the only linear combination of these vectors that equals to 0 is with $a_1 = a_2 = \dots = a_k = 0$.
 - Examples: $\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}, \left\{\begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}\right\}$
- A set of m -dimensional vectors $\{v_1, v_2, \dots, v_k\}$ is called **linearly dependent** if they are not linearly independent.
 - Examples: $\left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix}\right\}, \left\{\begin{bmatrix} 1 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}\right\}, \left\{\begin{bmatrix} 1 \\ 7 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 6 \\ 4 \end{bmatrix}\right\}$
- The maximum number of linearly independent columns (or rows) of a matrix is called the **rank** of that matrix.
 - Examples: $\text{rank} \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) = 2, \text{rank} \left(\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \right) = 1, \text{rank} \left(\begin{bmatrix} 1 & 1 & 6 \\ 7 & 2 & 4 \end{bmatrix} \right) = 2$

- A square $n \times n$ matrix A is called **non-singular or invertible** if it has n linearly independent columns (and rows), i.e., $\text{rank}(A) = n$.
 - An invertible matrix A has a unique **inverse** denoted by A^{-1} , which satisfies $AA^{-1} = A^{-1}A = I_n$.
 - Example: $\begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}^{-1} = \frac{1}{2} \begin{bmatrix} 8 & -2 \\ -3 & 1 \end{bmatrix}$.
- A square $n \times n$ matrix A is called **singular or non-invertible** if it has less than n linearly independent columns (and rows), i.e., $\text{rank}(A) < n$.
 - A singular matrix does not have a well-defined inverse.

Consider a system of m linear equations in n variables:

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m\end{aligned}$$

In compact matrix form, the system is written as $Ax = b$.

• Case 1: $m = n$

- $\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$
- $\begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 9 \end{bmatrix}$
- $\begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 10 \end{bmatrix}$

• Case 2: $m < n$

- $\begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \end{bmatrix}$
- $\begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$
- $\begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

• Case 3: $m > n$

- $\begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 8 \end{bmatrix}$
- $\begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}$
- $\begin{bmatrix} 2 & 1 \\ 4 & 2 \\ 6 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$

- Consider the linear system $Ax = b$ with m equations in n variables.
- It is convenient to write the system in **augmented form** $[A \mid b]$.
- We can use the **Gauss–Jordan method** (or Gaussian elimination) to conclude that the system has **no solution**, a **unique solution**, or **infinitely many solutions**.
- The idea is to apply **elementary row operations** to obtain an equivalent (triangular) system, and then use back-substitution.
- Each elementary row operation can be viewed as a matrix pre-multiplication.

• Scaling a row by a non-zero scalar:

• Example: $\begin{bmatrix} 2 & 1 & 3 \\ 4 & 8 & 12 \end{bmatrix} \xrightarrow{\frac{1}{4}\text{row}_2 \rightarrow \text{row}_2} \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 4 & 8 & 12 \end{bmatrix}$

• Multiplying a row with a non-zero scalar and adding to a different row:

• Example: $\begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 3 \end{bmatrix} \xrightarrow{-\frac{1}{2}\text{row}_1 + \text{row}_2 \rightarrow \text{row}_2} \begin{bmatrix} 2 & 1 & 3 \\ 0 & \frac{3}{2} & \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 3 \end{bmatrix}$

• Interchanging two rows:

• Example: $\begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & 3 \end{bmatrix} \xrightarrow{\text{row}_1 \leftrightarrow \text{row}_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & 3 \end{bmatrix}$

Consider the linear system $Ax = b$ and compare $\text{rank}(A)$ with $\text{rank}([A \mid b])$:

- If $\text{rank}(A) \neq \text{rank}([A \mid b])$: **no solution** (inconsistent system).
- If $\text{rank}(A) = \text{rank}([A \mid b]) = n$: a **unique solution**.
- If $\text{rank}(A) = \text{rank}([A \mid b]) < n$: **infinitely many solutions**.

- Suppose that we obtain **reduced row echelon form** of a matrix A , that is, we upper-triangularize A .
 - The number of nonzero elements along the diagonal gives the rank of A .
- If a square matrix A is of **full-rank**, we can obtain its inverse via row operations.
 - In particular, we start with the augmented matrix $[A \quad I]$ and apply row operations to obtain $[I \quad B]$. In this case, $B = A^{-1}$.

Consider $Ax = b$ with $m \leq n$ and $m = \text{rank}(A) = \text{rank}([A \mid b])$. Let B be an $m \times m$ **basis** of A (a set of m linearly independent columns) indexed by BV; the remaining variables are nonbasic (NBV).

- Set the nonbasic variables to 0.
- Solve $Bx_{\text{BV}} = b$ for the basic variables.
- The point $x_{\text{BV}} = B^{-1}b$, $x_{\text{NBV}} = 0$ is a **basic solution**.
- If in addition $x_{\text{BV}} = B^{-1}b \geq 0$, it is a **basic feasible solution (BFS)**.

- Consider the system $\begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$.
- Case 1:**
 - $BV=\{1,2\}$, $NBV=\{3,4\}$
 - $Bx_{BV} = b \leftrightarrow \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$
 - Note that $rank(B) = 2$ and $x_{BV} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -4 \\ 8 \end{bmatrix}$. Also, $x_{NBV} = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

- Consider the system $\begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$.
- **Case 2:**
 - $BV=\{1,3\}$, $NBV=\{2,4\}$
 - $Bx_{BV} = b \leftrightarrow \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$
 - Note that $rank(B) < 2$. No basic solution exists for this choice of BV.

- Consider the system $\begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$.
- Case 3:**
 - $BV=\{1,4\}$, $NBV=\{2,3\}$
 - $Bx_{BV} = b \leftrightarrow \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$
 - Note that $rank(B) = 2$ and $x_{BV} = \begin{bmatrix} x_1 \\ x_4 \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \end{bmatrix}$. Also, $x_{NBV} = \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

- Consider the system $\begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$.
- Case 4:**
 - $BV=\{2,3\}$, $NBV=\{1,4\}$
 - $Bx_{BV} = b \leftrightarrow \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$
 - Note that $rank(B) = 2$ and $x_{BV} = \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \end{bmatrix}$. Also, $x_{NBV} = \begin{bmatrix} x_1 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

- Consider the system $\begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$.
- Case 5:**
 - $BV=\{2,4\}$, $NBV=\{1,3\}$
 - $Bx_{BV} = b \leftrightarrow \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_2 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$
 - Note that $rank(B) < 2$. No basic solution exists for this choice of BV.

- Consider the system $\begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$.
- Case 6:**
 - $BV=\{3,4\}$, $NBV=\{1,2\}$
 - $Bx_{BV} = b \leftrightarrow \begin{bmatrix} 2 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$
 - Note that $rank(B) = 2$ and $x_{BV} = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$. Also, $x_{NBV} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if for all $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Examples: $f(x) = x^2$; $f(x_1, \dots, x_n) = a_1x_1 + \dots + a_nx_n = a^\top x$; $f(x) = e^x$. (Note $f(x) = x^3$ is **not** convex on $\mathbb{R}$.)

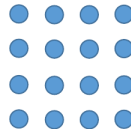
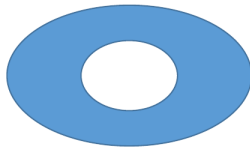
A function f is **concave** if $-f$ is convex. A **linear** function is both convex and concave.

- A set $S \subseteq \mathbb{R}^n$ is called **convex** if for all $x \in S, y \in S$ and $\lambda \in [0,1]$, we have $\lambda x + (1 - \lambda)y \in S$.

- Examples of **convex** sets:



- Examples of **non-convex** sets:



- Intersection of **convex** sets is a **convex** set but union or difference of **convex** sets is **not necessarily convex**.

- A **hyperplane** is defined as

$$\left\{ x \in \mathbb{R}^n : \sum_{j=1}^n a_j x_j = b \right\}.$$

- A **half-space** is defined as

$$\left\{ x \in \mathbb{R}^n : \sum_{j=1}^n a_j x_j \leq b \right\}.$$

- Note that **hyperplanes** and **half-spaces** are convex sets.
- Note that any **hyperplane** is the intersection of two **half-spaces**.
- The intersection of finitely many half-spaces is called a **polyhedron** (hence, a **polyhedron** is a convex set).
- A **bounded polyhedron** is called a polytope.

- In general, the following problems are "easy" to solve:
- minimizing a **convex** function over the intersection of convex sets;
- maximizing a **concave** function over the intersection of convex sets.
- In general, the following problems are "hard" to solve:
- minimizing a **concave** function;
- maximizing a **convex** function;
- optimizing a **non-convex** function (including problems with integrality restrictions).
- How about the difficulty of solving LPs?

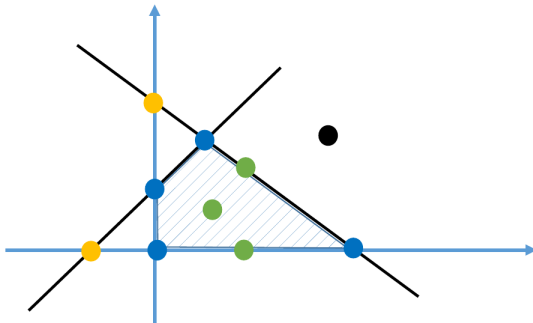
Recall the generic form of an LP:

$$\max; \sum_{j=1}^n c_j x_j \quad \text{s.t.} \quad \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m.$$

Notice that the **feasible region of any LP is a polyhedron**. Therefore, understanding the structure of polyhedra is essential to understanding some important properties of LPs.

- It is possible to solve LPs with two variables by the “Graphical Solution Method”.
- This method gives valuable intuition as to the **possible outcomes of LPs**:
 - LP is infeasible (no feasible solution exists)
 - LP is feasible (feasible solution exists)
 - LP has unbounded objective (the “optimal value” is unbounded)
 - LP has bounded objective (the optimal value is bounded)
 - Unique optimal solution
 - Multiple optimal solution
- We will now see how the properties of polyhedra play a role in identifying which of these outcomes are realized.

- Let $S \subseteq \mathbb{R}^n$ be a convex set. A point $x \in S$ is called an **extreme point of S** if it cannot be written as the midpoint of two other points from S .



Consider $Ax = b$, $x \geq 0$, with $m \leq n$ and $m = \text{rank}(A) = \text{rank}([A \mid b])$. Let B be a basis indexed by BV ($\text{rank}(B) = m$).

- Set the nonbasic variables x_{NBV} to 0 and solve $Bx_{\text{BV}} = b$.
- The point $x_{\text{BV}} = B^{-1}b$, $x_{\text{NBV}} = 0$ is a **basic solution**.
- If moreover $x_{\text{BV}} = B^{-1}b \geq 0$, it is a **basic feasible solution (BFS)**.

Theorem. A point in the feasible region of an LP is an **extreme point** if and only if it is a **basic feasible solution** of the LP.

- Thus extreme points (defined *geometrically*) are equivalent to basic feasible solutions (defined *algebraically*).
- We will see how this property is exploited in the Simplex Method.

- Consider the problem

$$\max 4x_1 + 3x_2$$

$$\text{s.t. } x_1 + x_2 \leq 40$$

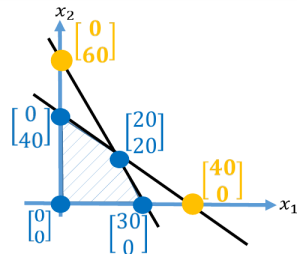
$$2x_1 + x_2 \leq 60$$

$$x_1, x_2 \geq 0$$

- Draw the feasible region. Identify the extreme points, basic solutions and basic feasible solutions.

Basic / Feasible Solutions and Extreme Points: Example 2

$$\begin{aligned} \max \quad & 4x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 40 \\ & 2x_1 + x_2 + x_4 = 60 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$



Basic Variables		Nonbasic Variables		Basic Solution		BFS/Extreme Point?	
x_1, x_2	x_3, x_4	$x_1 = 40, x_2 = 0, x_3 = 0, x_4 = 0$	YES	$x_1 = 40, x_2 = 0, x_3 = 0, x_4 = 0$	YES	YES	YES
x_1, x_3	x_2, x_4	$x_1 = 30, x_2 = 0, x_3 = 10, x_4 = 0$	YES	$x_1 = 30, x_2 = 0, x_3 = 10, x_4 = 0$	YES	NO	NO
x_1, x_4	x_2, x_3	$x_1 = 30, x_2 = 0, x_3 = 0, x_4 = 30$	YES	$x_1 = 30, x_2 = 0, x_3 = 0, x_4 = 30$	YES	NO	NO
x_2, x_3	x_1, x_4	$x_1 = 0, x_2 = 40, x_3 = 0, x_4 = 0$	YES	$x_1 = 0, x_2 = 40, x_3 = 0, x_4 = 0$	YES	YES	YES
x_2, x_4	x_1, x_3	$x_1 = 0, x_2 = 60, x_3 = 0, x_4 = 0$	YES	$x_1 = 0, x_2 = 60, x_3 = 0, x_4 = 0$	YES	NO	NO
x_3, x_4	x_1, x_2	$x_1 = 0, x_2 = 0, x_3 = 40, x_4 = 60$	YES	$x_1 = 0, x_2 = 0, x_3 = 40, x_4 = 60$	YES	NO	NO
x_1, x_2	x_3, x_4	$x_1 = 20, x_2 = 20, x_3 = 0, x_4 = 0$	YES	$x_1 = 20, x_2 = 20, x_3 = 0, x_4 = 0$	YES	YES	YES
x_1, x_3	x_2, x_4	$x_1 = 20, x_2 = 0, x_3 = 20, x_4 = 0$	YES	$x_1 = 20, x_2 = 0, x_3 = 20, x_4 = 0$	YES	NO	NO
x_1, x_4	x_2, x_3	$x_1 = 20, x_2 = 0, x_3 = 0, x_4 = 20$	YES	$x_1 = 20, x_2 = 0, x_3 = 0, x_4 = 20$	YES	NO	NO
x_2, x_3	x_1, x_4	$x_1 = 0, x_2 = 20, x_3 = 20, x_4 = 0$	YES	$x_1 = 0, x_2 = 20, x_3 = 20, x_4 = 0$	YES	YES	YES
x_2, x_4	x_1, x_3	$x_1 = 0, x_2 = 40, x_3 = 0, x_4 = 20$	YES	$x_1 = 0, x_2 = 40, x_3 = 0, x_4 = 20$	YES	NO	NO
x_3, x_4	x_1, x_2	$x_1 = 0, x_2 = 0, x_3 = 20, x_4 = 40$	YES	$x_1 = 0, x_2 = 0, x_3 = 20, x_4 = 40$	YES	NO	NO

- Consider the problem

$$\max x_1 + x_2 + x_3$$

$$\text{s.t. } x_1 - x_2 + x_3 + x_4 = 0$$

$$x_1 + x_3 + x_5 = 1$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

- Identify the basic solutions and basic feasible solutions.
- Given m equations in n unknowns, can you write a code which automatically generates **ALL** the basic (feasible) solutions?

Basic / Feasible Solutions and Extreme Points: Example 2

Basic Variables		Nonbasic Variables	Basic Solution				BFS/Extreme Point?
...	...	...	...	...	...	...	YES
...	...	...	...	...	...	...	N/A
...	...	...	...	...	...	...	NO
...	...	...	...	...	...	...	YES
...	...	...	...	...	...	...	YES
...	...	...	...	...	...	...	N/A
...	...	...	...	...	...	...	YES
...	...	...	...	...	...	...	NO
...	...	...	...	...	...	...	YES
...	...	...	...	...	...	...	YES

Recall that nonbasic variables are set to zero in a basic solution, whereas basic variables may take nonzero values.

If, at a basic solution, the value of **at least one basic variable is zero**, that basic solution is called **degenerate**.

In the case of degeneracy, multiple bases might represent the same point. For instance, the extreme point $x_1 = x_2 = x_3 = x_4 = x_5 = x_6 = 0$ can be identified by several different bases.

Let $S \subseteq \mathbb{R}^n$ be a nonempty convex set. A vector d is a **direction of unboundedness** (or recession direction) of S if $x + \lambda d \in S$ for all $x \in S$ and $\lambda \geq 0$.

A direction of unboundedness d is an **extreme direction** of S if it cannot be written as the midpoint of two other (distinct) directions of unboundedness of S .

Theorem. For a nonempty polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b, x \geq 0\}$, a vector $d \neq 0$ is a direction of unboundedness of P if and only if

$$Ad \leq 0, \quad d \geq 0.$$

Theorem. For a nonempty polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$, a vector $d \neq 0$ is a direction of unboundedness of P if and only if

$$Ad \leq 0.$$

- Consider the following standard form LP (assume it is feasible):

$$\begin{array}{ll} \max & \sum_{j=1}^n c_j x_j \\ \text{s.t.} & \sum_{j=1}^n a_{ij} x_j = b_i \quad i = 1, \dots, m \\ & x_j \geq 0 \quad j = 1, \dots, n. \end{array} \iff \begin{array}{ll} \max & c \cdot x \\ \text{s.t.} & Ax = b \\ & x \geq 0. \end{array}$$

- THEOREM:**

The optimal value of the above LP is bounded if and only if $c \cdot d \leq 0$ for all $d \in \mathbb{R}^n$ such that $Ad = 0, d \geq 0$.

- Consider the following canonical form LP (assume it is feasible):

$$\begin{array}{ll}
 \max & \sum_{j=1}^n c_j x_j \\
 \text{s.t.} & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m \\
 & x_j \geq 0 \quad j = 1, \dots, n.
 \end{array}
 \iff
 \begin{array}{ll}
 \max & c \cdot x \\
 \text{s.t.} & Ax \leq b \\
 & x \geq 0.
 \end{array}$$

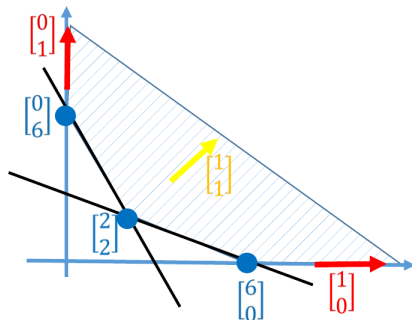
- THEOREM:**

The optimal value of the above LP is bounded if and only if $c \cdot d \leq 0$ for all $d \in \mathbb{R}^n$ such that $Ad \leq 0, d \geq 0$.

- For feasible maximization LPs, we can replace the condition " $c \cdot d \leq 0$ for all direction of unboundedness d " with the condition " $c \cdot d \leq 0$ for all the extreme directions d " in the previous **THEOREMS**.
- For feasible minimization LPs, we need to flip the sign in the conditions, that is, a minimization LP is bounded if and only if " $c \cdot d \geq 0$ for all the extreme directions d ".

- Find the extreme points and extreme directions of the following **minimization LP**:

$$\begin{aligned} \min \quad & c_1x_1 + c_2x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \geq 6 \\ & x_1 + 2x_2 \geq 6 \\ & x_1, x_2 \geq 0. \end{aligned}$$



- Extreme points: $\begin{bmatrix} 0 \\ 6 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$ Extreme directions: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\begin{aligned} \min \quad & c_1 x_1 + c_2 x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \geq 6 \\ & x_1 + 2x_2 \geq 6 \\ & x_1, x_2 \geq 0. \end{aligned}$$

Extreme directions: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

- The optimal value is **bounded** for $c = (1,1)$ since both $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are nonnegative.
- The optimal value is **unbounded** for $c = (-1,1)$ since $\begin{bmatrix} -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ or $\begin{bmatrix} -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is negative

$$\min c_1x_1 + c_2x_2$$

$$\text{s.t. } 2x_1 + x_2 \geq 6$$

$$x_1 + 2x_2 \geq 6$$

$$x_1, x_2 \geq 0.$$

Extreme directions: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

- For which of the following objective function coefficients, the optimal value is **bounded**? **Unbounded**?
 - $c = (1,2)$, $c = (2,1)$, $c = (0,1)$, $c = (1,0)$
 - $c = (-1,-1)$, $c = (1,-1)$

$$\min c_1 x_1 + c_2 x_2$$

$$\text{s.t. } 2x_1 + x_2 \geq 6$$

$$x_1 + 2x_2 \geq 6$$

$$x_1, x_2 \geq 0.$$

Extreme directions: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

- Specify the set of objective function coefficients for which the optimal objective value is **bounded**.
- The optimal value is **bounded** for $c = (c_1, c_2)$ if and only if both $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are nonnegative.
- Hence, we must have $c_1 \geq 0$ and $c_2 \geq 0$.

Theorem. Consider a nonempty polyhedron P (with at least one extreme point). Then any point $x \in P$ can be written as

$$x = \sum_{k=1}^K \lambda_k v^k + \sum_{l=1}^L \mu_l d^l, \quad \lambda_k \geq 0, \quad \sum_{k=1}^K \lambda_k = 1, \quad \mu_l \geq 0,$$

where $v^1, \dots, v^K$ are the **extreme points** of P and $d^1, \dots, d^L$ are the **extreme directions** of P .

- Recall the extreme points and extreme directions of the polyhedron:

$$2x_1 + x_2 \geq 6$$

$$x_1 + 2x_2 \geq 6$$

$$x_1, x_2 \geq 0.$$

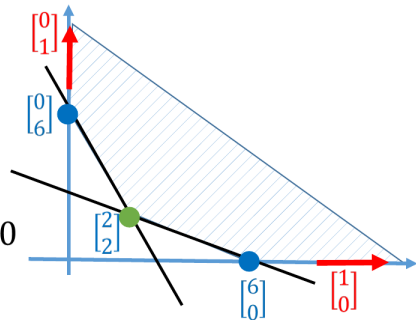
- Write the following points in terms of the extreme points and directions:
 - $x = (2,2), x = (6,0), x = (0,6), x = (3,3), x = (4,4), x = (8,7),$
 $x = (4,1), x = (1,1)$

Minkowski's Theorem: Illustration 2

$$\begin{aligned} \min \quad & c_1 x_1 + c_2 x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \geq 6 \\ & x_1 + 2x_2 \geq 6 \\ & x_1, x_2 \geq 0. \end{aligned}$$

- **Find** $\lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0, \mu_1 \geq 0, \mu_2 \geq 0$
with $\lambda_1 + \lambda_2 + \lambda_3 = 1$ such that

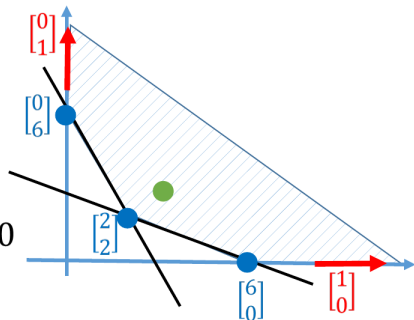
$$\begin{bmatrix} 2 \\ 2 \end{bmatrix} = \lambda_1 \begin{bmatrix} 0 \\ 6 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \lambda_3 \begin{bmatrix} 6 \\ 0 \end{bmatrix} + \mu_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \mu_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$



$$\begin{aligned}
 &\min c_1 x_1 + c_2 x_2 \\
 &\text{s.t. } 2x_1 + x_2 \geq 6 \\
 &\quad x_1 + 2x_2 \geq 6 \\
 &\quad x_1, x_2 \geq 0.
 \end{aligned}$$

- **Find** $\lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0, \mu_1 \geq 0, \mu_2 \geq 0$
with $\lambda_1 + \lambda_2 + \lambda_3 = 1$ such that

$$\begin{bmatrix} 3 \\ 3 \end{bmatrix} = \lambda_1 \begin{bmatrix} 0 \\ 6 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \lambda_3 \begin{bmatrix} 6 \\ 0 \end{bmatrix} + \mu_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \mu_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

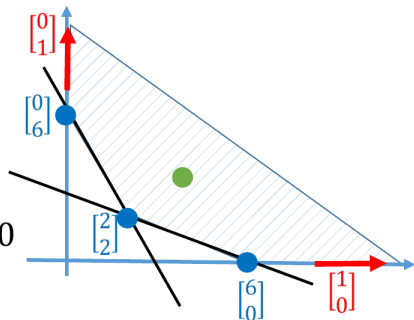


Minkowski's Theorem: Illustration 2

$$\begin{aligned} \min \quad & c_1 x_1 + c_2 x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \geq 6 \\ & x_1 + 2x_2 \geq 6 \\ & x_1, x_2 \geq 0. \end{aligned}$$

- **Find** $\lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0, \mu_1 \geq 0, \mu_2 \geq 0$
with $\lambda_1 + \lambda_2 + \lambda_3 = 1$ such that

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \lambda_1 \begin{bmatrix} 0 \\ 6 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \lambda_3 \begin{bmatrix} 6 \\ 0 \end{bmatrix} + \mu_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \mu_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$



- Extreme points: $\begin{bmatrix} 0 \\ 6 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$ Extreme directions: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- $\begin{bmatrix} 3 \\ 3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 \\ 6 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 6 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \begin{bmatrix} 0 \\ 6 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 6 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- $\begin{bmatrix} 4 \\ 4 \end{bmatrix} = 0 \begin{bmatrix} 0 \\ 6 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 6 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 8 \\ 7 \end{bmatrix} = 0 \begin{bmatrix} 0 \\ 6 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 6 \\ 0 \end{bmatrix} + 6 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- $\begin{bmatrix} 4 \\ 1 \end{bmatrix} = 0 \begin{bmatrix} 0 \\ 6 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 6 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 4 \\ 2 \end{bmatrix} = 0 \begin{bmatrix} 0 \\ 6 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 6 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is not feasible.

- Find the extreme points and extreme directions of the polyhedron:

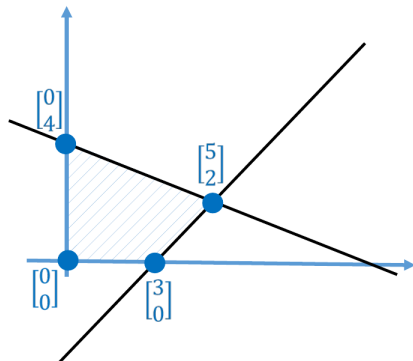
$$x_1 - x_2 \leq 3$$

$$2x_1 + 5x_2 \leq 20$$

$$x_1, x_2 \geq 0.$$

- Extreme points: $\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \end{bmatrix}$

- Extreme directions: $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ (feasible region is bounded)



- Recall the extreme points and extreme directions of the polyhedron:

$$x_1 - x_2 \leq 3$$

$$2x_1 + 5x_2 \leq 20$$

$$x_1, x_2 \geq 0.$$

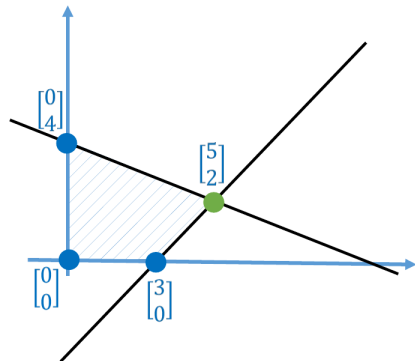
- Write the following points in terms of the extreme points and directions:
 - $x = (0,0)$, $x = (3,0)$, $x = (0,4)$, $x = (5,2)$, $x = (0,2)$, $x = (2,2)$,
 $x = (4,2)$, $x = (1,1)$, $x = (4,1)$, $x = (4,4)$, $x = (5,7)$

Minkowski's Theorem: Illustration 3

$$\begin{aligned}x_1 - x_2 &\leq 3 \\ 2x_1 + 5x_2 &\leq 20 \\ x_1, x_2 &\geq 0.\end{aligned}$$

- **Find** $\lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0, \lambda_4 \geq 0$
with $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1$ such that

$$\begin{bmatrix} 5 \\ 2 \end{bmatrix} = \lambda_1 \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 \\ 4 \end{bmatrix} + \lambda_3 \begin{bmatrix} 5 \\ 2 \end{bmatrix} + \lambda_4 \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$

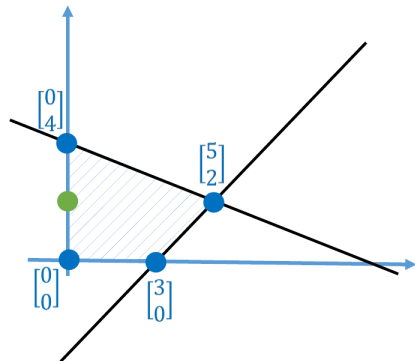


Minkowski's Theorem: Illustration 3

$$\begin{aligned}x_1 - x_2 &\leq 3 \\ 2x_1 + 5x_2 &\leq 20 \\ x_1, x_2 &\geq 0.\end{aligned}$$

- **Find** $\lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0, \lambda_4 \geq 0$
with $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1$ such that

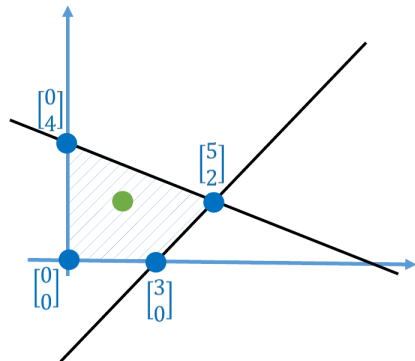
$$\begin{bmatrix} 0 \\ 2 \end{bmatrix} = \lambda_1 \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 \\ 4 \end{bmatrix} + \lambda_3 \begin{bmatrix} 5 \\ 2 \end{bmatrix} + \lambda_4 \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$



$$\begin{aligned}x_1 - x_2 &\leq 3 \\2x_1 + 5x_2 &\leq 20 \\x_1, x_2 &\geq 0.\end{aligned}$$

- **Find** $\lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0, \lambda_4 \geq 0$
with $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1$ such that

$$\begin{bmatrix} 2 \\ 2 \end{bmatrix} = \lambda_1 \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 \\ 4 \end{bmatrix} + \lambda_3 \begin{bmatrix} 5 \\ 2 \end{bmatrix} + \lambda_4 \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$



- Extreme points: $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 5 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$, Extreme directions: $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- $\begin{bmatrix} 0 \\ 2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 0 \\ 4 \end{bmatrix} + 0 \begin{bmatrix} 5 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 3 \\ 0 \end{bmatrix}$,
- $\begin{bmatrix} 2 \\ 2 \end{bmatrix} = \frac{3}{10} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \frac{3}{10} \begin{bmatrix} 0 \\ 4 \end{bmatrix} + \frac{2}{5} \begin{bmatrix} 5 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 3 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 4 \\ 2 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \frac{1}{10} \begin{bmatrix} 0 \\ 4 \end{bmatrix} + \frac{4}{5} \begin{bmatrix} 5 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 3 \\ 0 \end{bmatrix}$,
- $\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{5}{12} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 0 \\ 4 \end{bmatrix} + 0 \begin{bmatrix} 5 \\ 2 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 3 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 4 \\ 1 \end{bmatrix} = 0 \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 4 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 5 \\ 2 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 3 \\ 0 \end{bmatrix}$
- $\begin{bmatrix} 4 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ 7 \end{bmatrix}$ are not feasible.

Consider a feasible maximization LP. The Graphical Solution Method suggests that either the optimal value of the LP is **unbounded**, or there exists an **extreme point (BFS)** that is optimal.

- If $c^\top d^* > 0$ for some extreme direction d^* , then the LP is unbounded (we can move along d^* indefinitely while improving).
- Otherwise, an optimal solution is attained at an **extreme point (BFS)**.

For any feasible LP with m non-redundant constraints, two basic feasible solutions are **adjacent** if their sets of basic variables have $m - 1$ basic variables in common.

- Recall Example 2: the points A–B, A–C, B–D, and C–D are adjacent BFS (extreme point) pairs.
- What are the adjacent BFS pairs in Example 3?